

Reliability-Screened Routing Under Predictable and Stochastic Travel Variation: Matched Comparisons of Conservative Planning and Statistical Screening

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Abstract

Routes planned on nominal travel times can fail under predictable congestion and random delay. This paper audits a two-layer procedure for the capacitated vehicle-routing problem with time windows. A replaceable generator proposes plans, internal window contraction supplies candidate buffers without changing customer promises, and a plan is admitted only when a one-sided 95% Wilson lower bound on the probability that at least 95% of customers are served on time reaches 0.95. Selected plans are assessed once on untouched scenarios. On 53 seeded synthetic instances with 20 to 200 customers and OR-Tools candidates, the screened procedure exceeded uniformly conservative travel-time padding by 8.3–21.4 percentage points, and capping each customer’s contraction at its direct-reachability limit raised admission from 32 to 42 instances, at 2.5% more distance. Two timestamped confirmatory studies, each on 53 new instances from the same generator, followed. The first confirmed all four prespecified hypotheses: screening exceeded conservative padding by 15.7 points and capping added 11.7. The second replaced OR-Tools with hybrid genetic search. With HGS candidates, screening with capping exceeded HGS conservative padding by 27.9 points, and differed from the OR-Tools procedure by -0.2 points (95% interval -4.0 to 3.7). A conservative-speed fallback, suggested post hoc, did not measurably improve the capped procedure but helped the uncapped one. These results use the original objective weights; with equal weights the advantage is not distinguishable from zero. Pooled service probability stays below the 95% target, mainly through fallback plans. Results are conditional on assumed disturbance parameters and synthetic geometry.

Keywords: vehicle routing; time windows; stochastic travel times; chance constraints; simulation-based validation; reproducibility

MSC Classification: 90B06, 90C59, 90C15, 62G09

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1 Introduction

The vehicle-routing problem with time windows (VRPTW) asks a fleet to serve geographically distributed customers subject to capacity and scheduling constraints [16, 18]. In practice, a route that is feasible under one planning-time matrix may miss promised windows when congestion varies or small delays propagate downstream. Minimising nominal distance alone does not price that fragility.

This paper asks six questions:

1. Does the complete screened procedure improve on uniformly conservative planning when both are compared on identical instances?
2. Does any remaining reliability gain arise from route structure or simply from using more vehicles?
3. How do generator choice, clock-profile information and individual contraction caps change admission success and out-of-sample service?
4. How much distance, driver duration, and fleet use are exchanged for the resulting reliability?
5. Do the main findings repeat on fresh instances under a prespecified protocol, and how sensitive are they to the objective weights?
6. Does screening still improve on conservative planning when candidates come from a state-of-the-art generator, and does a conservative-speed fallback improve the procedure when screening admits nothing?

The proposed architecture separates *candidate generation* from *stochastic validation*. The optimizer is a nominal VRPTW solver and is not presented as an exact solver for a time-dependent stochastic program. Candidate routes are stress-tested after solving, first on a screening bank and then once on an untouched validation bank. The principal reliability event permits a small daily proportion of late deliveries, avoiding the mechanical collapse of an all-customers-on-time probability as instance size grows.

The contribution is integrative. Time-dependent speed profiles, multi-criteria routing, chance constraints, Monte Carlo evaluation, and Wilson intervals are established ideas. The paper contributes an architecture and the evidence for it. The architecture treats the routing solver as a replaceable candidate generator and makes every reliability decision after it: internal window contraction produces a family of candidates, a statistical screen admits or rejects each one, and an independent scenario bank validates the choice. Capping each customer’s contraction at its direct-reachability limit removes a failure mode in which contraction makes individually reachable customers unreachable. The evidence consists of a development study with ablations of uniform travel-time padding, clock-profile information, fleet resources, contracted candidates and screening; two confirmatory studies with publicly timestamped protocols; and a test in which the OR-Tools generator is replaced by hybrid genetic search. The screened family and the statistical selection rule act together; a contrast against one uncontracted plan does not identify the selection rule alone.

The study has three parts. The development study on 53 synthetic instances motivated the design, and its findings are exploratory. A first confirmatory study then tested four prespecified hypotheses on 53 new instances, and a further exploratory analysis varies the objective weights. A second confirmatory study, on another 53 new instances, replaced the OR-Tools generator with hybrid genetic search [20, 22] and tested the conservative-speed fallback that the first study had suggested post hoc.

2 Related work

Solomon’s benchmark study established the standard VRPTW experimental setting [16]; Toth and Vigo provide a broad treatment of vehicle-routing models and algorithms [18]. Time-dependent routing requires care because a naive multiplier applied at departure can violate first-in-first-out (FIFO). Ichoua, Gendreau, and Potvin introduced the integrated speed-profile construction used here [10]; Ghiani and Guerriero later analysed its representational properties [9].

Stochastic VRPTW research has treated uncertain travel and service times using chance constraints, recourse, approximate route-feasibility tests, and data-driven formulations [5, 8, 11, 15]. Ehmke, Campbell, and Urban embed per-customer service levels in a routing method [7]; Taş et al. jointly model time dependence, stochastic travel, service costs, and soft windows [17]; and Bomboi, Buchheim, and Prunte develop route-feasibility checks for correlated and time-dependent travel times [5]. Vareias, Repoussis, and Tarantilis optimize self-imposed time windows under stochastic travel times [19], which is the closest conceptual analogue to the internal window contraction used here.

The present method differs in where the statistical claim is made. It does not embed a stochastic feasibility approximation inside a specialist route-search algorithm. Instead, it accepts routes from a replaceable nominal solver, searches an internal-buffer family on one scenario bank, and audits the chosen plan once on a held-out bank. Its daily service event constrains the proportion of customers served on time across the whole day; per-customer chance constraints are not used. This solver-agnostic separation is narrower than a new stochastic optimizer, but makes selection failure and out-of-sample reliability directly observable. Monte Carlo approximations to chance constraints require separate assessment of candidate feasibility [1]; the Wilson bound is used because normal intervals behave poorly near probabilities close to one [6, 21].

The implementation uses Google OR-Tools as a nominal candidate generator [13]. It is a practical local-search framework and gives no optimality guarantee. PyVRP provides a substantially stronger hybrid genetic search (HGS) for canonical VRPTW variants [20, 22]. It is used here as a deterministic benchmark and, in the second confirmatory study, as an alternative candidate generator.

3 Problem definition

Let $N = \{0, 1, \dots, n\}$ be the nodes, with depot 0 and customers $C = N \setminus \{0\}$. Let V be a homogeneous fleet. Customer i has demand δ_i , deterministic service time s_i , and offered service-start window $[l_i, u_i]$. Vehicle capacity is Q , and the maximum shift duration is $H^{\max}$. The nominal distance and planning time on arc (i, j) are d_{ij} and $\bar{t}_{ij}$.

The planning variables are $x_{ijv} \in \{0, 1\}$ for arc use, $z_{iv} \in \{0, 1\}$ for customer assignment, $y_v \in \{0, 1\}$ for vehicle use, service-start time a_{iv} , depot departure d_v , depot return r_v , and slack deficit $G_{iv} \geq 0$. Assignment variables explicitly activate the customer–vehicle copies:

$$\sum_{v \in V} z_{iv} = 1 \quad i \in C, \quad (1)$$

$$\sum_{j \in N \setminus \{i\}} x_{ijv} = \sum_{j \in N \setminus \{i\}} x_{jiv} = z_{iv} \quad i \in C, v \in V, \quad (2)$$

$$\sum_{j \in C} x_{0jv} = \sum_{i \in C} x_{i0v} = y_v \quad v \in V. \quad (3)$$

For delivery-only demand, route capacity can be stated without cumulative-load variables:

$$\sum_{i \in C} \delta_i z_{iv} \leq Q y_v \quad v \in V. \quad (4)$$

The time-propagation constraints exclude customer-only subtours: summing them around any directed customer cycle would imply $0 \geq \sum_i (s_i + \bar{t}_{ij}) > 0$. Positive demands are not needed for that argument. Time windows apply only to assigned copies:

$$l_i z_{iv} \leq a_{iv} \leq u_i z_{iv} \quad i \in C, v \in V. \quad (5)$$

Using separate departure and return variables avoids representing two depot visits with one time variable. With a valid global deactivation constant $M \geq H^{\max} + \max_i s_i + \max_{ij} \bar{t}_{ij}$,

$$a_{jv} \geq d_v + \bar{t}_{0j} - M(1 - x_{0jv}), \quad (6)$$

$$a_{jv} \geq a_{iv} + s_i + \bar{t}_{ij} - M(1 - x_{ijv}), \quad (7)$$

$$r_v \geq a_{iv} + s_i + \bar{t}_{i0} - M(1 - x_{i0v}). \quad (8)$$

The bounds $0 \leq d_v, r_v \leq H^{\max} y_v$ deactivate unused vehicles, and $r_v - d_v \leq H^{\max} y_v$. The implementation enforces these timing relations through OR-Tools Routing dimensions and never builds (8) explicitly, so no big- M relaxation or big- M diagnostic is claimed for the code.

For target slack s^* ,

$$G_{iv} \geq a_{iv} - (u_i - s^*) z_{iv}, \quad (9)$$

$$G_{iv} \geq 0. \quad (10)$$

This makes $G_{iv} = \max\{0, s^* - (u_i - a_{iv})\}$ when $z_{iv} = 1$ and zero when the customer is not assigned to vehicle v .

4 Candidate objectives and scaling

The four raw criteria are

$$F_1 = c_f \sum_{v \in V} \sum_{i,j \in N} d_{ij} x_{ijv}, \quad F_2 = \sum_{v \in V} \sum_{i \in C} G_{iv}, \quad (11)$$

$$F_3 = \sum_{v \in V} (r_v - d_v), \quad F_4 = \sum_{v \in V} y_v. \quad (12)$$

Thus F_3 includes driving, service, and waiting while permitting each vehicle to depart when useful. This removes the fixed-departure artifact that charged a vehicle for depot waiting before its first feasible service.

Single-criterion heuristic solves provide empirical reference values. For each criterion k , let L_k and U_k be the smallest and largest values observed across these anchor solutions. These values are *scaling ranges*, not utopia and nadir points, and they need not contain the Pareto frontier. No value is clipped. The normalized score is

$$Z(\mathbf{x}) = \sum_{k=1}^4 w_k \frac{F_k(\mathbf{x}) - L_k}{\max\{U_k - L_k, 1\}}. \quad (13)$$

The unit fallback prevents division by zero when the anchor archive shows no variation in a discrete criterion. The baseline weights $\mathbf{w} = (0.5693, 0.2643, 0.1055, 0.0609)$ are the principal-eigenvector weights of a single Analytic Hierarchy Process (AHP) pairwise-comparison matrix with consistency ratio 0.0254 [14]; the matrix and calculation appear in Appendix A and in `code/ahp_weights.py`. The matrix was specified for an earlier criteria set in which the third criterion was total distance, and its third row and column were not re-elicited when that criterion became driver duration. The weights are therefore a fixed experimental preference vector that defines the candidate-selection objective. They were fixed before any experiment reported here,

were not elicited from a fleet operator, and make no claim about the preferences of any industry. Section 10.8 reports a sensitivity analysis over four weight vectors.

OR-Tools implements the distance term as an arc cost, driver duration as a per-vehicle Time-dimension span cost, slack deficit as customer soft upper bounds, and fleet use as a fixed vehicle cost. Integer coefficients are scaled once at the criterion level. The primary planning matrix is nominal and fixed; the original candidate generator does not use clock-dependent costs. The *conservative-speed* baseline replaces 40 km/h by 33.3 km/h throughout the planning matrix. It pads all trips by 20%, including off-peak trips; it does not isolate knowledge of the peak’s timing.

4.1 Clock-aware approximation and deterministic guarantee

A separate distance heuristic freezes each origin’s estimated departure from the preceding schedule, integrates (14) at that time to construct an outgoing travel-time row, and re-solves. Two passes receive 2.5 seconds each. The first reference is the retained nominal plan, whose separate solve cost has already been incurred. Customer departures are estimated service start plus service time; for depot arcs, the estimate is the preceding departure for the first customer of each route, or the mean of used depot departures for other customers. Each resulting sequence is rescheduled with the exact deterministic FIFO profile. Binary search finds its latest feasible whole-minute depot departure; routes failing at departure zero are rejected. The shortest deterministically feasible output is retained, with duration breaking ties. This is an approximation with explicit failure reporting, not a departure-dependent callback or an exact optimizer for the time-dependent VRPTW.

For a feasible conservative-speed plan, the evaluation speed at $\sigma = 0$ never falls below the planning speed during the planned shift. Starting from the same depot departure, FIFO and induction along each route imply that evaluated service starts and return occur no later than their planned times. Thus deterministic 100% on-time service is guaranteed by construction. The clock-aware approximation’s deterministic feasibility is likewise checked before acceptance. These properties are not experimental reliability gains.

5 Time-dependent stochastic evaluation

5.1 FIFO travel times

Partition the day into periods $[T_k, T_{k+1})$ with strictly positive speeds ν_k . The deterministic time-dependent travel time $t_{ij}(\tau)$ is the smallest $\Delta \geq 0$ satisfying

$$\int_{\tau}^{\tau+\Delta} \nu(u) du = d_{ij}, \quad \nu(u) = \nu_k \text{ for } u \in [T_k, T_{k+1}). \quad (14)$$

Because distance accumulated by time is strictly increasing, the arrival map $\tau \mapsto \tau + t_{ij}(\tau)$ is nondecreasing and FIFO holds.

5.2 Correlated disturbances

For scenario ω , arc travel is

$$T_{ij}(\tau, \omega) = t_{ij}(\tau) \Omega_{ij}(\omega), \quad (15)$$

where

$$\Omega_{ij}(\omega) = \exp\{\varepsilon_{ij}(\omega)\}, \quad (16)$$

$$\varepsilon_{ij}(\omega) = \sigma \left(\sqrt{\rho} \xi_0(\omega) + \sqrt{1-\rho} \xi_{ij}(\omega) \right). \quad (17)$$

Here ρ is the correlation between latent Gaussian log-errors. It is not the Pearson correlation between multipliers, which equals

$$\text{Corr}(\Omega_i, \Omega_j) = \frac{\exp(\sigma^2 \rho) - 1}{\exp(\sigma^2) - 1}. \quad (18)$$

The multiplier has median 1 and mean $\exp(\sigma^2/2)$, so the deterministic profile represents the median travel condition, which lies slightly below the mean.

Scenario banks are generated independently of route iteration order. A common factor is drawn once per scenario, while each directed arc receives a deterministic pseudorandom stream keyed by the bank seed and arc endpoints. Plans evaluated on the same bank therefore experience identical disturbances on shared arcs.

6 Screening and independent validation

For a fixed plan R and scenario ω , let

$$q(R, \omega) = \frac{1}{|C|} \sum_{i \in C} \mathbf{1}\{a_i(R, \omega) \leq u_i\} \quad (19)$$

be the on-time customer proportion. The primary service event is

$$J_\kappa(R, \omega) = \mathbf{1}\{q(R, \omega) \geq \kappa\}, \quad \kappa = 0.95, \quad (20)$$

and the target is

$$\Pr[J_\kappa(R, \omega) = 1] \geq \alpha, \quad \alpha = 0.95. \quad (21)$$

The stricter zero-lateness fleet-day probability is still reported, but is not the sole admission condition because it declines mechanically as the number of customers increases.

For N scenarios and empirical success $\hat{p}$, the one-sided Wilson lower bound at confidence $1 - \eta$ is

$$L(\hat{p}) = \frac{\hat{p} + z^2/(2N) - z\sqrt{\hat{p}(1 - \hat{p})/N + z^2/(4N^2)}}{1 + z^2/N}, \quad z = z_{1-\eta}. \quad (22)$$

Candidate generation contracts only the internal planning deadline: $[l_i, \max\{l_i, u_i - \beta\}]$. The offered window $[l_i, u_i]$ is unchanged, and every scenario is scored against the original u_i . Candidate values of β are generated with the slack-aware objective and assessed on a screening bank. The anchor ranges are computed once from the uncontracted ($\beta = 0$) instance and held fixed over the buffer search. Among candidates whose screening lower bound reaches α , the one with the smallest normalized planned score (13) is selected. In simulation, every vehicle departs at its planned d_v ; there is no scenario-dependent departure recourse. The selected plan is evaluated once on a larger, untouched validation bank. This separation prevents adaptive candidate search from selecting on the evidence used for final reporting.

For procedure-level reporting, a prespecified fallback makes the output defined on every instance. If no candidate passes screening, the uncontracted slack-aware plan ($\beta = 0$) is returned. Validation never changes this choice; it only estimates the performance of the plan already chosen. Selected-only results are shown only when labelled as such; the primary procedure comparison includes all instances.

Three reliability quantities appear in the results, and they answer different questions:

- **Admission criterion.** The Wilson lower bound (22) on the screening bank, computed for each candidate. A candidate is admitted when it reaches $\alpha = 0.95$.
- **Selected-plan probability.** The validated estimate of $\Pr[J_\kappa = 1]$ for a plan that passed screening, from the untouched bank.

- **Procedure-level probability.** The validated probability averaged over all instances, including those on which screening admitted nothing and the fallback was returned.

The chance constraint (21) is the requirement that screening tests for each admitted plan. The procedure-level average falls below α whenever fallback plans are weaker, so the results report admitted plans and the procedure as a whole separately.

6.1 Generator replacement and reachability-capped contraction

Screened conservative-speed uses the same buffer grid, admission bank and Wilson threshold, but generates distance-minimizing plans with the padded matrix. Among passing plans it chooses the shortest distance, breaking ties by planned duration and then buffer. Its fallback is the uncontracted conservative plan; if that plan was not found, the retained uncontracted slack-aware plan is used. This makes the replacement procedure defined on all 53 instances while preserving the base generator’s 46/53 found-plan count.

Let $e_i = \max\{l_i, \bar{t}_{0i}\}$ be earliest direct service under the nominal matrix. The capped weighted variant uses the internal upper bound

$$u_i^{\text{cap}}(\beta) = \min\{u_i, \max\{l_i, u_i - \beta, e_i\}\}. \quad (23)$$

This prevents an initially reachable customer from becoming unreachable by a direct trip solely because of contraction. It does not imply joint routing feasibility or the admission target. Identical contracted models reuse their archived candidates; only changed models are re-solved. Selection still uses the fixed original scaling ranges and weighted score, with the same fallback.

For diagnosis, a shortest-path relaxation uses arc cost $\bar{t}_{ij} + s_i$ (with $s_0 = 0$), allowing intermediate customers without capacity or window restrictions. Its depot-to- i distance is a valid lower bound even if rounded distances violate a triangle inequality. A contracted deadline below this bound certifies infeasibility. Without a certificate, a missing solver output is labelled *no feasible plan found*; neither a time limit nor passing the reachability test proves feasibility. Solver status certificates were not retained in the archived runs and are not inferred retrospectively.

7 Experimental design

Synthetic instances place a central depot and uniformly distributed customers on a 100×100 grid. Demand is uniform on 10–49 units, service time is 10 minutes, window width is uniform on 90–149 minutes, capacity is 1000, and the shift limit is 600 minutes. Time zero is 06:00 local time, so simulated clock time is obtained by adding 360 minutes to each relative schedule time. The retained design uses the seeded sizes in Table 1; all model and simulation constants are disclosed in Table 2.

Table 1 Seeded synthetic experimental design.

Customers	Vehicles available	Instances	Seeds
20	6	20	7200–7219
50	14	15	7500–7514
100	26	10	8000–8009
200	40	8	9000–9007

Eight configurations are compared:

1. **Nominal distance:** fixed 40 km/h planning matrix and a distance-minimizing objective.

Table 2 Planning, disturbance, and statistical parameters.

Parameter	Retained value
Nominal/free-flow speed $\bar{v}$	40 km/h
Shift and clock origin	600 min, beginning 06:00 local time
Capacity; service time; target slack s^*	1000; 10 min; 15 min
Distance cost c_f	15 cost units/km
AHP weights (w_1, w_2, w_3, w_4)	(0.5693, 0.2643, 0.1055, 0.0609)
Integer objective scale	100,000
Planned-shift speed divisors	0.9 (06:00–07:00), 1.2 (07:00–09:00), and 1.0 (09:00–16:00); speed is $\bar{v}$ divided by the divisor
Outside-shift continuation	1.0 to 17:00, 1.3 (17:00–19:00), 0.9 thereafter; used if a stochastic route overruns the shift
Log-error scale σ ; latent correlation ρ	0.20; 0.50
Exploratory disturbance grid	$\sigma \in \{0.10, 0.20, 0.30\}$; $\rho \in \{0, 0.5, 0.9\}$
Service fraction κ ; target α	0.95; 0.95
Wilson tail probability η	0.05 (one-sided 95% lower bound)
Screening; validation; sensitivity scenarios	1000; 5000; 1000 per cell
Buffer grid	$\beta \in \{0, 5, \dots, 60\}$ min

2. **Conservative-speed distance:** distance minimization with a uniform 33.3 km/h planning matrix.
3. **Slack-aware:** objective (13) with $\beta = 0$, isolating the objective from screening.
4. **Fleet-matched distance:** distance minimization forced to use exactly the fleet count of the slack-aware plan and initialized from its feasible routes.
5. **Original procedure:** the complete weighted internal-buffer search, with selection on 1000 screening scenarios, the prespecified $\beta = 0$ fallback, and assessment on 5000 fresh scenarios.
6. **Screened conservative:** conservative-speed distance candidates with the same admission test and the fallback defined above.
7. **Capped procedure:** the original weighted procedure with customer-specific reachability caps.
8. **Clock approximation:** two departure-estimate matrix updates and exact deterministic route rescheduling.

The nominal shift ends at 16:00. The full-day speed profile remains defined beyond that time because lognormal delays are unbounded and a stochastic route can overrun. Evening periods lie outside the planned shift but remain in the evaluator. Fleet-matched distance is initialized from the slack-aware routes; because this can in principle transfer their structure, the initialization is treated as part of the comparator’s definition.

The screening grid is $\beta \in \{0, 5, \dots, 60\}$ minutes. The four single-criterion anchor solves use two seconds each to stabilize the scaling ranges; reported candidate configurations use a five-second OR-Tools search budget for each nominal solve. If a candidate solve returns no feasible plan, the same five-second call is attempted at most twice more; the first feasible result is kept, and no best-of-several selection is made. Reachability-certified new models need no solver attempt. The clock approximation instead divides a total five-second budget between its two solves. Six experiment workers ran in parallel; elapsed time and software metadata are retained. All configurations use the same fresh validation bank within an instance. The

reported reliability outcomes are service-event probability, zero-latency fleet-day probability, mean on-time fraction, expected number late, mean and 95th-percentile total latency, and mean route success. Operational outcomes are distance, driver duration, and fleet use. Simulated duration is additionally reported under the common evaluation model, to distinguish actual route time from each generator’s planned F_3 . Differences are paired by instance. Confidence intervals across instances are distinguished from binomial Monte Carlo intervals within an instance. Across-instance intervals use a paired nonparametric bootstrap with 20,000 resamples and a fixed analysis seed. Claims of distance equivalence require a pre-specified equivalence margin; failure to reject a difference is not treated as proof of equality.

8 First confirmatory study and weight-sensitivity design

A confirmatory protocol was fixed before any fresh instance was generated or solved. It specifies four directional hypotheses, their tests and decision rules, a failure policy, and 53 instance seeds equal to the development seeds shifted by 60000. The protocol, the code that runs and analyses it, and a SHA-256 manifest of those files were committed to a public repository (commit 41fae67d9520bd5dd01fcac87e680a5a69325d10) and archived on Zenodo [2] before the study was run. The analysis program refuses to run unless the complete prescribed study is present and every frozen file matches the manifest.

The hypotheses concern validated service-event probability on the fresh bank, with the instance as the unit. H1 (primary): on instances where conservative-speed planning returns a plan, the original procedure exceeds it. H2 (primary): the capped procedure exceeds the original procedure on all instances. H3: the capped procedure is admitted by screening on more instances than the original procedure, tested with an exact McNemar test at 0.05. H4: on the instances of H1, the advantage of the original procedure over conservative speed is larger at $\sigma = 0.30$ than at $\sigma = 0.10$ with $\rho = 0$. H1 and H2 are decided at the 97.5% level, a Bonferroni adjustment for two primary hypotheses, and are confirmed only if their paired bootstrap interval lies above zero. H4 uses a 95% interval. The two other hypotheses cannot rescue an unconfirmed primary one. H3 and H4 are secondary and carry less evidential weight than H1 and H2, and Holm’s procedure in place of the Bonferroni adjustment would not change the two primary decisions. H4 compares fixed plans: each plan is screened under the primary disturbance setting ($\sigma = 0.20$, $\rho = 0.5$) and then evaluated across the sweep without re-screening. A configuration without a plan scores zero on that instance, in validation and in every sweep cell, so a failure counts against the configuration that failed. A software error is recorded with its traceback, is never scored as routing performance, and blocks the analysis.

Parameters, budgets and scenario counts equal those of the development study (Table 2). The study ran on an Apple M1 Pro laptop with ten cores under macOS 26.3.1, with Python 3.12.15, NumPy 2.5.3 and OR-Tools 9.15.6755. The development runs used Python 3.12.14 on Linux. Stage 1 launched the four instance sizes as four parallel processes where the written protocol lists a sequential loop; stage 2 used six workers as specified. These are the deviations from the written procedure. The scheduling change affects only the CPU time available to each time-limited solve.

The weight analysis is exploratory. The original procedure was rerun on the development instances with four weight vectors: the original vector, one that swaps its two largest weights, a distance-heavy vector and equal weights (Section 10.8). Each rerun regenerates the candidate family, screens it on the same bank and validates the selected plan or fallback on the same fresh bank, with the same five-second budget. The vectors were fixed before the runs, and each ran once.

The benchmark check is also exploratory. It uses the eight R1 and RC1 Solomon files available in the ML4VRP 2024 competition repository [12], which hosts Solomon’s instances [16]. The C1 instances are excluded because their horizon of 1236 time units exceeds one working

shift. Coordinates are read as kilometres, with the study’s rounded Euclidean distances and a planning speed of 40 km/h, so that one Solomon time unit equals 1.5 minutes. Ready and due times are multiplied by 1.5 and floored, service time becomes 15 minutes, and the capacity of 200 and the 25 vehicles are kept. The shift length is the scaled depot due time, 345 minutes for R1 and 360 minutes for RC1, starting at 06:00 so that the morning peak falls inside the shift. Planning times are rounded up to whole minutes as in the study. The mapping keeps Solomon’s relation between travel time and window width, because one distance unit takes one Solomon time unit to drive at the planning speed. The C1 instances could be fitted by compressing time, but that would change the relative tightness of their windows, so the eight instances form a convenience sample of the R1 and RC1 types. The pipeline code is unchanged; only the instance loader and the parameter constructor are replaced.

9 Second confirmatory study design

The first confirmatory study left two questions open. Its candidate generator is below the state of the art in route quality, so the benefit of screening might shrink when candidates come from a stronger search. Its post hoc analysis also suggested that returning the conservative-speed plan when screening admits nothing would raise pooled service (Section 10.7), a rule chosen after the results were seen. A second protocol tested both questions on 53 further instances whose seeds are the development seeds shifted by 80000.

The second generator is the hybrid genetic search of PyVRP [20, 22], version 0.14.0. It solves the same planning model as the OR-Tools generator: the static matrix at the planning speed, hard and possibly contracted windows, capacity, shift length and fleet size, with each vehicle departing at the latest feasible time. Its weighted objective keeps the distance, driver-duration and fleet terms with the same coefficients. PyVRP cannot express the slack-deficit term, so that term is left out of generation and kept in candidate selection. Each solve stops after 10,000 iterations with the instance seed as its random seed. The scaling ranges come from the unchanged OR-Tools anchor solves and are shared by both generators; because those solves are time-limited, the HGS half is deterministic given the ranges. The contraction grid, capped windows, reachability certificates, scenario banks, admission rule and selection score are those of the OR-Tools procedure. HGS conservative-speed planning solves the distance objective at 33.3 km/h. The OR-Tools half reruns the frozen pipeline of the first study unchanged.

Two fallback rules are compared. The slack-aware rule is the one of the first study; for HGS it returns the uncontracted weighted plan. The conservative rule returns the conservative-speed plan of the same generator when screening admits nothing, and the slack-aware plan when no conservative plan exists. Neither rule changes a screening decision or uses a validation outcome.

The protocol fixes four hypotheses on validated service-event probability, with the instance as the unit. G1 (primary): on instances where HGS conservative-speed planning returns a plan, the HGS capped procedure with conservative fallback exceeds it. G2 (primary): on all instances, the conservative fallback improves the OR-Tools capped procedure. G3 (secondary): the same holds for the uncapped OR-Tools procedure. G4: the difference between the HGS and OR-Tools capped procedures, both with conservative fallback, is estimated with a 95% interval and no directional decision. G1 and G2 are decided at the 97.5% level and G3 at 95%, with the bootstrap, failure policy and decision rules of the first study. The protocol requires the abstract and conclusion to report an unconfirmed primary hypothesis.

The protocol, the HGS code, the analysis program and a SHA-256 manifest covering them and the frozen files of the first study were committed to the public repository (commit 3ba6b71288a2a51872ab6a877fa86eb28ff7671d) and archived on Zenodo [3] before any instance of the study was solved. A gate script checked the manifest, the pushed commit and tag, and the absence of results before the runs began. An earlier draft of the protocol prescribed seeds shifted by 70000. The OR-Tools half was run on those seeds before that draft was timestamped,

and its console output, which lists the instances screening admitted, was seen. That seed set was abandoned in full: its records are kept in the archive and enter no analysis, the protocol discloses them, and the study used the seeds shifted by 80000 with no other change. The HGS generator was built and tested on the development instances, so the development HGS results are exploratory. The study ran on the machine and software of the first confirmatory study, with PyVRP 0.14.0, and followed the written procedure without deviation.

10 Results

10.1 Complete procedures on matched instances

Table 3 puts every original configuration on the same conservative-feasible subset. The paired contrasts are computed from individual instance differences. Their intervals therefore quantify the comparison directly; they are not obtained by subtracting endpoints of separate confidence intervals.

Table 3 Validated service-event probability (%) on the same 46 instances: every entry in a row has the displayed denominator. Entries are mean [95% instance-bootstrap interval]. N, C, S, F and P denote nominal, conservative-speed, slack-aware, fleet-matched and original procedure, respectively.

n	k	N	C	S	F	P
20	18	21.5 [18.8, 24.4]	64.2 [61.1, 67.4]	53.3 [47.6, 59.4]	21.7 [18.9, 24.8]	85.7 [74.4, 95.0]
50	13	17.7 [16.1, 19.6]	59.0 [54.6, 63.1]	45.8 [43.1, 48.4]	16.9 [14.8, 18.9]	76.8 [62.3, 89.7]
100	8	25.4 [23.5, 27.1]	71.6 [69.1, 74.0]	55.3 [51.5, 59.3]	26.2 [23.6, 29.0]	87.6 [74.2, 97.5]
200	7	26.7 [24.9, 28.8]	74.6 [72.2, 77.0]	61.7 [58.3, 66.8]	27.7 [25.5, 30.6]	82.9 [69.3, 94.3]

Table 4 Paired probability differences (percentage points) on exactly the same subsets as Table 3. SC denotes screened conservative-speed with its prespecified fallback.

n	S minus C	P minus C	SC minus C
20	-11.0 [-16.1, -6.0]	21.4 [9.6, 31.3]	28.2 [21.4, 34.3]
50	-13.1 [-17.4, -8.9]	17.9 [2.8, 32.0]	11.8 [2.7, 22.2]
100	-16.2 [-20.6, -11.7]	16.0 [4.8, 24.8]	11.2 [2.8, 19.4]
200	-12.9 [-18.5, -5.7]	8.3 [-5.1, 20.0]	5.7 [0.0, 12.9]

In the paired contrasts of Table 4, the complete original procedure improves on conservative-speed planning by 8.3–21.4 percentage points across sizes. The unscreened slack-aware objective does not show this advantage. At $n = 200$ the interval includes zero, so the positive mean at that size is not conclusive.

10.2 Resources on the common subset

Scheduled driver duration is not identical to realized labour time: the conservative planner budgets padded arc times, while the evaluator may traverse those arcs faster. Table 5 therefore includes both measures. Absolute distance premiums also carry solver-quality uncertainty: the five-second deterministic OR-Tools benchmark remains about 11% longer than HGS at $n = 200$.

The matched data do not support uniform resource dominance: at $n = 20$ the original procedure has slightly greater scheduled time and more vehicles; at $n = 100$ it also uses slightly more vehicles. The simulated-time interval excludes zero only for the increase at $n = 20$. The service gain is therefore a trade-off against some operating resources.

Table 5 Paired original-procedure changes relative to conservative speed on the common subset. Distance and duration are mean percentage changes; fleet is an absolute difference. Scheduled duration is each planner’s own F_3 . Simulated duration uses the same realized traffic model and planned departures for both configurations.

n	Distance	Scheduled time	Simulated time	Fleet
20	4.9 [2.4, 7.3]	0.3 [-3.8, 4.8]	8.5 [4.4, 12.9]	0.22 [0.00, 0.44]
50	2.3 [-0.6, 5.2]	-7.5 [-12.2, -2.7]	-2.0 [-7.0, 2.8]	-0.46 [-1.00, 0.00]
100	2.0 [-1.3, 5.1]	-3.0 [-9.4, 2.5]	2.2 [-4.3, 7.6]	0.12 [-0.38, 0.62]
200	2.4 [-1.4, 5.7]	-7.7 [-15.2, -1.0]	-2.8 [-10.6, 4.1]	-0.71 [-2.57, 1.00]

10.3 Screening a replaceable generator and capping contractions

Table 6 All-instance procedure outcomes. Selected counts admission-bank lower bounds reaching 0.95; retained counts those selections whose fresh validation lower bound also reaches 0.95. Probability includes every fallback, as well as selected plans.

n	Procedure	Selected / instances	Retained / selected	Probability (%)
20	Original procedure	14/20	14/14	81.6 [70.2, 91.9]
20	Screened conservative	15/20	15/15	87.7 [79.0, 94.7]
20	Capped procedure	14/20	14/14	81.6 [70.3, 92.0]
50	Original procedure	8/15	8/8	72.7 [59.2, 86.4]
50	Screened conservative	4/15	4/4	67.5 [58.2, 77.3]
50	Capped procedure	10/15	10/10	80.1 [66.9, 90.8]
100	Original procedure	6/10	5/6	81.3 [68.8, 93.3]
100	Screened conservative	4/10	4/4	77.5 [67.2, 88.2]
100	Capped procedure	10/10	9/10	97.5 [96.7, 98.2]
200	Original procedure	4/8	3/4	80.6 [68.7, 92.3]
200	Screened conservative	2/8	2/2	78.3 [71.0, 86.5]
200	Capped procedure	8/8	7/8	97.6 [96.9, 98.2]

In Table 6, the screened conservative generator selects on 25/53 instances; 25 selections retain the target lower bound. Its all-instance paired difference from the original procedure is -0.3 [-5.5, 5.0] points. The capped variant selects on 42/53, with 40 retained; its corresponding difference is 7.7 [3.5, 12.5] points. Capping removes only an individual reachability obstruction; capacity, sequencing and finite-search limitations remain.

Table 7 Operational price of the capped variant relative to the original procedure on all 53 instances. Percentage changes are paired per instance; fleet changes are absolute. Capping can permit different route and resource choices, so its reliability gain is not attributed to schedule structure alone.

n	Distance	Scheduled time	Simulated time	Fleet
20	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.00 [0.00, 0.00]
50	2.2 [0.0, 5.4]	4.6 [0.0, 11.4]	4.5 [0.0, 11.3]	0.33 [0.00, 0.87]
100	4.9 [1.1, 8.9]	6.8 [1.7, 12.1]	6.6 [1.7, 11.8]	0.30 [-0.20, 0.80]
200	6.3 [1.6, 10.9]	10.5 [3.1, 18.0]	10.0 [2.9, 17.1]	1.88 [0.50, 3.38]

Pooled over all instances (Table 7), the capped variant costs 2.5% more distance and 0.43 additional vehicles than the original procedure. At $n = 200$, the additional fleet use is material and should not be obscured by the earlier fleet-matched ablation of the uncontracted objective.

Table 8 Clock-aware approximation: identical matched subsets within each row, restricted to an exact deterministic schedule being found by the approximation. Probabilities and paired changes are percentages and percentage points, respectively. The two matrix solves share a five-second search allowance.

n	Found	Nominal	Clock-aware	Original P	Clock minus N
20	19/20	21.2 [18.6, 24.0]	39.5 [37.9, 41.3]	84.1 [73.2, 93.6]	18.4 [15.1, 21.4]
50	14/15	17.5 [16.0, 19.3]	36.5 [35.4, 37.7]	74.8 [60.8, 87.1]	19.0 [17.2, 20.9]
100	10/10	25.4 [23.9, 26.8]	44.5 [43.0, 46.0]	81.3 [68.8, 93.2]	19.1 [17.6, 20.8]
200	6/8	26.6 [24.5, 28.9]	43.7 [42.3, 45.3]	81.7 [68.7, 93.6]	17.2 [15.7, 18.7]

10.4 Clock-aware approximation

The clock-aware rows test a practical departure-time approximation. Exact deterministic retiming verifies each reported route, so its success at $\sigma = 0$ is an acceptance condition and says nothing about stochastic robustness. Conservative-speed planning likewise guarantees deterministic on-time service within the planned horizon by upper-bounding every arc time. Both 100% rows hold by construction.

In Table 8, the approximation improves on nominal distance by 17.2–19.1 points on its feasible subsets, but remains below the original procedure. Its latest-feasible-departure rule and two-pass search are part of this result; the comparison does not establish what an optimal time-dependent planner could achieve.

10.5 Matched disturbance sensitivity

Table 9 Fixed-plan sensitivity on the common 46 instances. Each cell uses 1,000 common scenarios from a bank separate from primary validation; plans are not re-screened or reoptimized for each parameter setting. C, P and SC are conservative speed, original procedure and screened conservative.

σ	ρ	C (%)	P (%)	SC (%)	P minus C (points)
0.10	0.0	91.8 [88.2, 95.0]	88.8 [81.3, 95.3]	97.1 [94.3, 99.2]	-3.1 [-10.7, 3.9]
0.10	0.5	84.6 [81.5, 87.4]	87.8 [81.5, 93.5]	93.2 [90.1, 95.9]	3.2 [-3.3, 9.3]
0.10	0.9	81.2 [78.3, 84.0]	87.6 [81.4, 93.3]	91.5 [88.2, 94.5]	6.4 [0.0, 12.5]
0.20	0.0	69.8 [64.1, 75.5]	81.0 [71.4, 89.7]	86.7 [81.0, 92.0]	11.2 [1.4, 20.6]
0.20	0.5	65.6 [63.2, 67.9]	82.8 [75.9, 89.2]	82.9 [78.0, 87.5]	17.2 [10.3, 23.7]
0.20	0.9	67.2 [65.3, 68.9]	82.5 [76.8, 87.9]	82.1 [77.9, 86.2]	15.4 [9.5, 20.9]
0.30	0.0	41.7 [37.4, 46.0]	72.1 [61.1, 82.4]	69.4 [60.6, 77.8]	30.5 [19.8, 40.5]
0.30	0.5	54.4 [52.5, 56.3]	74.0 [67.8, 79.8]	72.1 [67.1, 77.0]	19.6 [13.6, 25.3]
0.30	0.9	60.6 [59.3, 61.9]	75.6 [70.7, 80.1]	74.4 [70.6, 78.2]	15.0 [10.0, 19.6]

The mechanism comparison of interest is the paired P-minus-C column of Table 9, which Fig. 1 also plots across the disturbance grid. Changes in its sign or magnitude describe fixed-plan performance under each setting. On its own, this column cannot separate screening from the contracted candidate family, and the nine cells share instances and common random numbers.

For independent disturbances, increasing σ from 0.10 to 0.30 changes the paired contrast from -3.1 [-10.7, 3.9] to 30.5 [19.8, 40.5] points. This supports a growing advantage of the complete screened family in that regime. At $\rho = 0.9$ the increase is not monotonic, so the data do not support a universal monotonic relationship between noise scale and screening value.

10.6 Diagnosis of screening failures

Of the 21 original screening failures, 17 reach a buffer with an explicit customer-reachability contradiction and 4 have no such certificate through 60 minutes. 8 instances also contain a no-plan outcome before any certificate. Those outcomes remain unresolved: failure of a necessary

reachability test proves infeasibility, but passing it does not establish a feasible multi-customer routing. The per-instance audit appears in Appendix B.1.

An additional exploratory check took the 8 failures whose search ended at a no-plan outcome without a reachability certificate and re-solved exactly that contracted model with a longer budget: 30 s, then 120 s for those still without a plan. In 4 cases a plan was found at that contraction level, so the original outcome was a search limit. The remaining 4 still have no plan after 120 s, which does not prove infeasibility. The script and records are part of Online Resource 1.

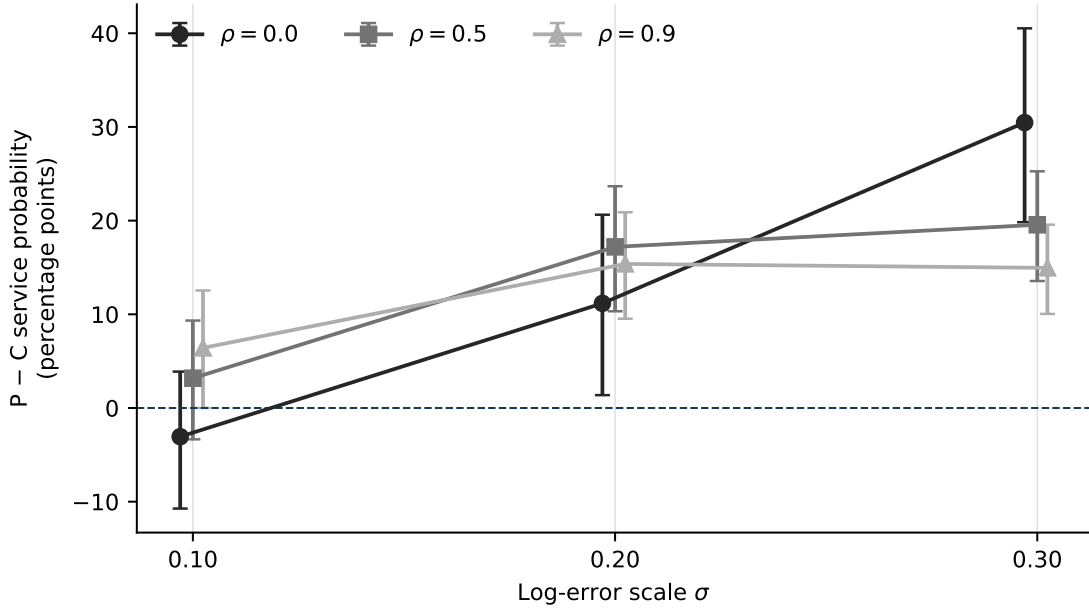


Fig. 1 Paired difference in service probability between the original procedure (P) and conservative-speed planning (C) on the common 46 instances. Circles, squares and triangles mark $\rho = 0$, 0.5 and 0.9, and the dashed line marks zero. Faint vertical lines mark $\sigma = 0.10$, 0.20 and 0.30; markers are offset slightly around them for legibility only. Error bars are 95% bootstrap intervals over instances. Plans are fixed across this disturbance sweep, and the intervals are not simultaneous across cells

10.7 First confirmatory study

The four hypotheses of the prespecified protocol were evaluated on 53 new instances (Table 10). Every decision uses the unrounded interval limits. The protocol’s validation passed before any quantity was computed.

All four hypotheses were confirmed. The original procedure exceeded conservative speed by 15.7 points, capping exceeded the original procedure by 11.7 points, and capping admitted 46 instances against 32. Capping lost no instance that the original procedure admitted (14 gained, 0 lost). The advantage over conservative speed was 28.3 points larger at $\sigma = 0.30$ than at $\sigma = 0.10$ for independent disturbances. Of the plans admitted on the screening bank, 31 of 32 (original) and 45 of 46 (capped) retained the 0.95 lower bound on the fresh bank. Pooled over all instances, service probability was 79.4% for the original procedure, 82.4% for screened conservative speed and 91.0% for the capped procedure, so the 95% target is still not met on average. Screened conservative speed was not a prespecified comparison, and its pooled value is descriptive only. As in the development study, the shortfall comes from instances without an admitted plan. Plans admitted by screening averaged 97.4% for the original procedure and

Table 10 Confirmatory results on 53 fresh instances. Differences are in percentage points of validated service-event probability; intervals are paired bootstrap intervals over instances. Conservative speed returned a plan on 50 of 53 instances, and H1 and H4 use those instances as the protocol defines.

Hypothesis	Quantity	Estimate [interval]	Level	Decision
H1 (primary)	Original procedure minus conservative speed, 50 instances	15.7 [7.5, 23.6]	97.5%	confirmed
H2 (primary)	Capped minus original procedure, 53 instances	11.7 [6.0, 17.9]	97.5%	confirmed
H3	Admitted by capped / original procedure	46 / 32 (capped only 14, original only 0)	exact $p = 0.00012$	confirmed
H4	Advantage at $\sigma = 0.30$ minus at $\sigma = 0.10$ ($\rho = 0$), 50 instances	28.3 [19.6, 36.4]	95%	confirmed

97.5% for the capped procedure, while the fallback plans averaged 51.9% (21 instances) and 48.7% (7 instances).

H2 and H3 are not independent. When neither procedure admits a plan, both return the same fallback, and 11.6 of the 11.7 points of H2 come from the 14 instances that only the capped procedure admits. The two results are one finding measured twice.

Post hoc fallback. The protocol fixes the fallback: without an admitted plan, both procedures return the uncontracted slack-aware plan. A variant that returns the conservative-speed plan instead, where that plan exists, would have raised pooled service on the same fresh bank from 79.4% to 85.0% for the original procedure (18 instances changed) and from 91.0% to 93.0% for the capped procedure (6 instances changed). The fallback rule uses no validation outcome, but this variant was chosen after the results were seen, so it is exploratory. The second confirmatory study tests it prospectively (Section 10.10).

These instances come from the same generator and the same assumed disturbance model as the development study. The result establishes repeatability within that setting. It does not address real road networks, calibrated parameters or other instance families.

10.8 Sensitivity to the objective weights

The original procedure was rerun on the 53 development instances with four weight vectors (Table 11); the confirmatory study used the original vector only. Each rerun regenerates the candidate family, screens it on the same bank and validates the selected plan or fallback on the same fresh bank. Conservative-speed planning does not use the weights, so its archived outcome is reused. The rerun of the original weights differs from the archived run by -1.4 points pooled ($[-5.1, 2.2]$) and by one admitted instance (31 against 32), which is the run-to-run variation of a time-limited search.

Swapping the two largest weights changed nothing beyond the run-to-run variation. The distance-heavy vector gave a lower pooled value, by 3.2 points, with a paired interval that includes zero, and its advantage over conservative speed stayed clearly positive (12.7 [5.3, 19.7]). Equal weights gave the largest change. They admitted 23 instances against 31, lowered pooled service by 8.8 points (paired interval $-8.8 [-14.3, -3.5]$; -15.6 to -2.4 after adjusting the level for three comparisons), and left an advantage over conservative speed whose interval includes zero (6.3 $[-1.1, 13.7]$). Of the instances whose admission differed, 9 were admitted only under the original weights and 1 only under equal weights (exact $p = 0.021$, unadjusted; the adjusted threshold for three comparisons is 0.017).

Table 11 Weight sensitivity on the 53 development instances. Weights are ordered fuel, slack, driver duration, fleet use. The advantage over conservative speed uses the 46 instances where it returned a plan. The last column is the paired difference in service probability from the rerun of the original weights over all 53 instances (percentage points).

Vector	Weights	Admitted	Pooled (%)	Advantage over conservative [95%]	Paired difference [95%]
Original (rerun)	(0.5693, 0.2643, 0.1055, 0.0609)	31	77.5	16.2 [9.2, 22.8]	–
Slack first	(0.2643, 0.5693, 0.1055, 0.0609)	30	77.4	16.0 [9.6, 22.1]	–0.1 [–3.9, 3.3]
Distance heavy	(0.8, 0.1, 0.05, 0.05)	29	74.3	12.7 [5.3, 19.7]	–3.2 [–7.1, 0.2]
Equal	(0.25, 0.25, 0.25, 0.25)	23	68.7	6.3 [–1.1, 13.7]	–8.8 [–14.3, –3.5]

This analysis is exploratory. It uses the development instances and four vectors chosen without a search for the point at which the advantage fades, and each vector was run once. The records do not store route structure, so the reason equal weights perform worse is not tested here. Equal weights give fleet size and driver duration a weight of 0.25 each, against 0.06 and 0.11 in the original vector.

10.9 Solomon instances

The unchanged pipeline was run on eight 100-customer Solomon instances with the time-dependent profile and disturbance model of the study overlaid (Section 8). Within its five-second budget, OR-Tools returned a nominal plan for five of the eight instances (Table 12). PyVRP/HGS finds a feasible plan for the other three, R101, R102 and RC105, within 40 seconds, so these outcomes are search limits of the generator and not infeasibility of the mapped instances. The failure policy records them as missing plans, and every dependent configuration is recorded as not attempted.

Table 12 Solomon instances: validated service-event probability (%). A dash marks a configuration with no plan. *Plan* states whether OR-Tools returned a nominal plan; *Admitted* states whether screening admitted a plan for the original procedure, which otherwise returns the slack-aware fallback.

Instance	Plan	Nominal	Conservative	Slack-aware	Original	Capped	Admitted
R101	no	–	–	–	–	–	no
R102	no	–	–	–	–	–	no
R103	yes	14.5	–	45.1	45.1	45.1	no
R112	yes	29.4	78.0	53.7	99.4	99.4	yes
RC104	yes	22.8	82.0	68.9	97.7	97.7	yes
RC105	no	–	–	–	–	–	no
RC106	yes	19.9	65.0	42.1	42.1	42.1	no
RC108	yes	28.3	80.7	51.5	98.8	98.8	yes

On the five instances with a plan, screening admitted a plan for three (R112, RC104 and RC108). Their validated service probability was 97.7 to 99.4%, against 78.0 to 82.0% for conservative speed and 22.8 to 29.4% for nominal distance. On the other two, the procedure returned the slack-aware fallback: R103 (45.1%; conservative speed returned no plan there, although HGS finds a feasible plan for its matrix) and RC106 (42.1%), where it fell 22.9 points below conservative speed (65.0%). Capping changed no result on any instance.

On RC106 the procedure was therefore 22.9 points worse than conservative speed. Screening admitted no plan there, and the slack-aware fallback is weaker than the conservative plan on that instance. The conservative-speed fallback examined post hoc in Section 10.7 would have returned 65.0%.

The eight instances are all the R1 and RC1 files available in the source repository. They give no basis for an interval estimate, and no Solomon comparison was prespecified. The admitted plans reached 97.7 to 99.4% and the fallback plans 42.1 to 45.1%, a split similar to that on the synthetic instances. The generator, however, returned no plan on a substantial share of these harder instances.

10.10 Second confirmatory study

In the exploratory development run on the 53 development instances, the HGS capped procedure with conservative fallback exceeded HGS conservative-speed planning by 27.0 points and pooled 94.2%. Table 13 gives the prespecified results on the 53 fresh instances of the second study.

Table 13 Second confirmatory study on 53 fresh instances. Differences are in percentage points of validated service-event probability; intervals are paired bootstrap intervals over instances. G1 uses the 49 instances where HGS conservative-speed planning returned a plan.

Hypothesis	Quantity	Estimate [interval]	Level	Decision
G1 (primary)	HGS capped with conservative fallback minus HGS conservative, 49 instances	27.9 [23.0, 32.8]	97.5%	confirmed
G2 (primary)	OR-Tools capped, conservative minus slack-aware fallback, 53 instances	0.4 [−0.7, 2.1]	97.5%	not confirmed
G3	OR-Tools uncapped, conservative minus slack-aware fallback, 53 instances	3.8 [1.6, 6.2]	95%	confirmed
G4	HGS minus OR-Tools, both capped with conservative fallback, 53 instances	−0.2 [−4.0, 3.7]	95%	estimate only

G1 was confirmed: with the HGS generator, screening with capped contraction exceeded conservative padding by 27.9 points, and pooled service probability reached 91.5% against 58.4% for HGS conservative speed. Capping was essential with this generator: at 100 and 200 customers the HGS capped family was admitted on 18 of 18 instances and the uncapped family on 3.

G2 was not confirmed. The OR-Tools capped procedure admitted a plan on 45 of 53 instances and fell back on 8. On 4 of those no conservative-speed plan existed, so both rules returned the same plan. On the remaining 4 the conservative fallback changed service probability by +5.4, −3.1, −12.0 and +30.0 points. The post hoc gain reported for the capped procedure in the first study did not replicate. For the uncapped procedure, which falls back more often, the conservative fallback helped (G3, 3.8 points).

The difference between the two generators under the capped procedure with conservative fallback (G4) was −0.2 points with a 95% interval from −4.0 to 3.7. No equivalence margin was prespecified, so this interval bounds the difference without establishing equivalence. The HGS $\beta = 0$ plan, which lacks the slack term, was a weak fallback: the uncapped HGS procedure pooled 51.0% with it and 74.7% with the conservative fallback (descriptive).

Appendix B reports the results of both confirmatory studies by size (Tables 14 and 15, descriptive only), customer outcomes on the matched subset (Table 16), absolute planned resources (Table 17), service-threshold sensitivity (Table 18), the instance-level failure audit (Table 19), and the equal-budget deterministic benchmark against PyVRP/HGS (Table 20).

11 Reproducibility and implementation checks

The study archive (Online Resource 1) contains the candidate generator, scenario-bank implementation, test suite, experiment driver, raw JSON records, and the script that produces the manuscript tables and figure. Tests cover FIFO behavior, lognormal moments, order-invariant common random numbers, Wilson calculations, visit and capacity feasibility, and the requirement that contracted candidates are scored against original windows. Runtime metadata record Python, NumPy, OR-Tools, operating system, solver budget, seeds, and scenario counts.

The clock convention is essential for reproduction: relative time zero is 06:00 and the evaluator adds 360 minutes before entering the full-day speed profile. Omitting that offset instead evaluates a 00:00–10:00 shift. The vectorized evaluator is checked against the original scalar implementation including period boundaries, customer lateness and shared arc disturbances.

The revision protocol was fixed before running the new candidates. Admission uses seed 100000+ instance seed; all fixed outputs are assessed with 5000 fresh scenarios from seed 400000+ instance seed. The 1000-scenario disturbance sweep uses seed 500000+ instance seed. Earlier aggregate results motivated these analyses, so they form an exploratory extension on known synthetic instances. They are neither preregistered nor an external replication. The fresh draws do not remove reuse of the instance set. Plans in the sweep are not re-screened when σ or ρ changes. The confirmatory protocol of Section 8 was publicly timestamped with a hash manifest of the code that runs and analyses it before the fresh instances were solved, and its results are reported in Section 10.7. One pipeline smoke test, on an instance outside the confirmatory set, was run while the protocol was prepared and is disclosed in the protocol as pilot data. The confirmatory study used only frozen scripts. The weight-sensitivity analysis, run after the freeze, uses a script outside the frozen file list. A first run of the weight-sensitivity script stopped on a coding error that affected only that script, and the corrected script produced the reported results. The second confirmatory study (Section 9) has its own frozen protocol and manifest. Its OR-Tools half uses the frozen scripts of the first study, and its HGS half is deterministic given the scaling ranges. The script that writes its tables and text stops unless its figures equal the saved output of the frozen analysis.

Every reported table and figure is regenerated from retained machine-readable records. Internal planning buffers are always evaluated against offered windows, candidate selection and final assessment use independent banks, and solver-quality comparisons use equal wall-clock budgets and identical deterministic instances. These rules prevent implementation diagnostics or adaptive within-run selection from using its own validation outcomes.

Use of generative AI tools. Generative AI assistants were used for code development, experiment implementation, code review, analysis checking, and drafting and editing text. The assistants were OpenAI’s ChatGPT (Astra 6) and Anthropic’s Claude (Opus 5, Opus 5.5 and Sonnet 5). They are not authors. Every experiment was run from the archived code, every reported number regenerates from the archived raw records, and the headline results were recomputed independently of the analysis scripts. Rerunning the solver can return different routes, because each solve stops at a wall-clock limit; Section 10.8 quantifies this run-to-run variation. The author reviewed all code and text and takes full responsibility for the content.

12 Limitations

First, the generators use nominal or iteratively frozen matrices, so the procedure can miss better stochastic plans. The OR-Tools generator is also below the state of the art: under an equal five-second budget, it returns routes about 11% longer than PyVRP/HGS at 200 customers (Table 20). Absolute distance premiums therefore carry solver quality, and some of the uncertified screening failures may reflect search limits. The second confirmatory study shows

that the reliability result does not depend on that generator, but its HGS generator omits the slack-deficit term, so the two generators do not solve identical objectives. Conservative speed is uniform padding, and the clock-aware approximation is not an exact time-dependent optimizer. Generator and objective choices also differ between screened families; the experiment is not a full factorial decomposition of every design decision. Second, the disruption parameters are assumed and have not been calibrated to fleet GPS residuals, so the results hold conditionally on those values and are not operational forecasts. An exploratory σ - ρ grid is reported, but it does not replace local calibration. Third, the geometry is synthetic. The Solomon check covers only 8 instances with 100 customers, and the generator found plans for 5 of them. The deterministic solver check is not a substitute for Gehring–Homberger benchmarks or a real road network. The confirmatory instances share the generator and the disturbance model of the development study, so they test repeatability within that setting. Fourth, the weights come from one preference vector. Under equal weights the advantage over conservative speed is not distinguishable from zero (Section 10.8), only four vectors were tried, and the confirmatory study used the original vector alone. Fifth, Wilson screening addresses binomial sampling uncertainty for a fixed candidate; its nominal coverage is not a simultaneous guarantee across an adaptively searched grid. Held-out assessment is therefore necessary. It is also not a proof of feasibility under distributional misspecification. Sixth, breakdown recovery, heterogeneous fleets, new-order insertion, and rolling-horizon replanning are excluded from the present evidence. So is departure-time recourse: every vehicle leaves at its planned time in every scenario, which makes the evaluation stricter than an operator who adjusts departures would face. Seventh, the fallback remains weak. The slack-aware plan can be worse than conservative padding when screening admits nothing (Section 10.9), and the conservative-speed fallback improved the uncapped procedure in the second confirmatory study but not measurably the capped one, which falls back on few instances.

13 Conclusion

The evidence separates three mechanisms. Uniformly conservative travel times protect against predictable congestion by construction, padding every trip including off-peak ones. On matched development instances the complete screened procedure exceeded them by 8.3–21.4 percentage points on average, although the interval at 200 customers includes zero, and for independent disturbances its advantage grew with noise, from -3.1 to 30.5 points. The unscreened slack-aware objective alone falls below conservative padding. The advantage comes from the complete procedure, in which the contracted candidate family and statistical selection act together.

A first confirmatory study on 53 fresh instances, with hypotheses fixed in a timestamped protocol before any of them was solved, confirmed all four prespecified hypotheses with the original weight vector. The screened procedure exceeded conservative speed by 15.7 points (97.5% interval 7.5 to 23.6), capping exceeded the original procedure by 11.7 points (6.0 to 17.9), capping admitted 46 instances against 32 with none lost, and the advantage over conservative speed grew with independent noise. These instances share the generator and the assumed disturbance model with the development study, so the result shows repeatability within that setting.

The advantage depends on the objective weights. Swapping the two largest weights changed nothing beyond run-to-run variation, and a distance-heavy vector kept a clearly positive advantage. With equal weights, fewer instances were admitted (23 against 31) and the advantage over conservative speed fell to 6.3 points with an interval from -1.1 to 13.7 . The confirmatory study used the original weights only.

On 8 Solomon instances with the same overlay, OR-Tools found a plan for 5 (PyVRP/HGS finds feasible plans for the other 3), and screening admitted 3 of those 5. The admitted plans reached service probabilities near 98%, and the fallback plans stayed below 46%. This check was

exploratory, covers a small set of instances, and shows that the candidate generator limits what the pipeline can do on harder benchmarks.

The second confirmatory study replaced OR-Tools with hybrid genetic search. With that generator, screening with capped contraction exceeded conservative-speed planning by 27.9 points (97.5% interval 23.0 to 32.8), and pooled service reached 91.5% against 58.4% for conservative speed. The HGS and OR-Tools procedures differed by -0.2 points, with a 95% interval from -4.0 to 3.7 ; no equivalence margin was set, so the interval bounds the difference without showing equivalence. The reliability gain is therefore not an artefact of a weak generator. Capping mattered more with HGS: at 100 and 200 customers its uncapped family was admitted on 3 of 18 instances and its capped family on 18.

The fallback is the weakest part of the procedure. When screening admits nothing, the original protocol returns the uncontracted slack-aware plan, which on RC106 was far worse than conservative speed (Section 10.9). A conservative-speed fallback, examined post hoc, would have raised pooled service in the first confirmatory study to 85.0% for the original procedure and 93.0% for the capped procedure. Tested prospectively in the second study, it did not measurably improve the capped procedure (G2, 0.4 points, interval -0.7 to 2.1): that procedure fell back on only 8 instances, and on 4 of them the rule changed the plan, with mixed effects. It did improve the uncapped procedure, by 3.8 points (95% interval 1.6 to 6.2).

Screening failures can be diagnosed. Among the 21 development instances without an admitted plan, 17 encounter a certified reachability obstruction at one or more tested contraction levels. A certificate proves that a particular contracted model is infeasible; it does not show that reachability alone caused the procedure to miss its target. Capping each customer’s contraction at its direct-reachability limit raised admission from 32 to 42 instances. Of the instances with 100 or 200 customers, 18/18 were admitted and 16/18 retained the target lower bound under fresh validation, with mean service probability 97.5–97.6%. The gain costs 2.5% more distance and 0.43 more vehicles on average, rising to 1.88 more vehicles at 200 customers. The earlier fleet-matched experiment concerns the uncontracted objective and does not isolate the capped variant’s mechanism. At 20 customers capping changes nothing; only two of the six failures there encounter a certificate. Of the other four, a longer search budget finds a plan for two, and two remain unresolved.

Pooled over all instances, no procedure reaches the 95% target, so procedure averages and selected-plan compliance must both be reported. In the development study, plans admitted by screening average 97.3% for the original procedure and 97.4% for the capped procedure, while the fallback plans used elsewhere average 50.9% and 45.2%. All 11 capped-procedure fallbacks occur at 20 or 50 customers, so closing the remaining gap depends on admitting plans for small instances. In the confirmatory study the capped procedure reached 91.0% pooled, against 79.4% for the original procedure. These results are conditional on assumed disruption parameters and synthetic geometry. The development findings were exploratory, and the two confirmatory studies support the main ones. Evidence from real road networks, other benchmark families and calibrated disturbance parameters remains outstanding.

Statements and Declarations

Competing interests. The author has no relevant financial or non-financial interests to disclose.

Funding. No funding was received for conducting this study.

Data and code availability. The reproducibility archive contains source code, dependency versions, tests, seeds, raw JSON records, analysis scripts, generated tables and figures, and both confirmatory protocols. It is supplied as Online Resource 1 and archived on Zenodo [4]. The

two confirmatory protocol freezes, each archived before any instance of its study was solved, are separate records [2, 3]. The Solomon instance files in the archive are third-party benchmark data [12, 16] and remain under their original terms.

Use of generative AI. Described in Section 11.

Ethics approval. Not applicable: the study uses synthetic data only.

Consent. Not applicable.

Author contributions. S.A. is the sole author and conceived the study, implemented the software, ran the experiments, and wrote the manuscript.

A AHP pairwise-comparison matrix

Criteria are ordered as distance cost, slack deficit, driver duration and fleet use. Entry a_{kl} is the judged importance of criterion k relative to l on Saaty’s 1–9 scale:

$$\mathbf{A} = \begin{pmatrix} 1 & 3 & 5 & 7 \\ 1/3 & 1 & 3 & 5 \\ 1/5 & 1/3 & 1 & 2 \\ 1/7 & 1/5 & 1/2 & 1 \end{pmatrix}.$$

Its principal eigenvalue is $\lambda_{\max} = 4.0685$, and the normalised principal eigenvector is

$$\mathbf{w} = (0.5693, 0.2643, 0.1055, 0.0609).$$

With Saaty’s random index 0.90 for four criteria, the consistency index is $(4.0685 - 4)/3 = 0.0228$ and the consistency ratio is 0.0254. As stated in Section 4, the third row and column were set when the third criterion was total distance and were not re-elicited for driver duration.

B Additional outcomes and failure records

Table 14 First confirmatory study by size (descriptive). For each procedure, the number of instances admitted by screening, the number of those that retained the 0.95 lower bound on the fresh bank in parentheses, and the mean validated service-event probability (%) over all instances including fallbacks.

n	Inst.	Original		Screened conservative		Capped	
20	20	17/20 (17)	90.2	15/20 (15)	87.8	17/20 (17)	90.2
50	15	8/15 (8)	74.6	8/15 (8)	80.3	12/15 (12)	87.4
100	10	4/10 (3)	71.3	3/10 (3)	80.0	10/10 (9)	97.1
200	8	3/8 (3)	71.5	2/8 (2)	75.4	7/8 (7)	92.4

Table 15 Second confirmatory study by size (descriptive): instances admitted by the capped procedure and mean validated service-event probability (%) over all instances, including fallbacks.

n	Inst.	OR-Tools capped			HGS		
		Admitted	Slack fb.	Cons. fb.	Admitted	Conservative	Capped, cons. fb.
20	20	13	83.8	83.3	14	56.8	85.8
50	15	14	93.7	95.7	13	61.9	92.4
100	10	10	97.9	97.9	10	64.1	97.4
200	8	8	97.5	97.5	8	48.9	97.1

Table 16 Customer outcomes on the conservative-feasible subset; the clock approximation is restricted further when needed. Counts make this additional conditioning explicit.

Configuration	k	On time (%)	Expected late
Nominal	46	83.6 [82.5, 84.6]	10.00 [7.95, 12.20]
Conservative speed	46	93.9 [93.4, 94.4]	3.47 [2.84, 4.14]
Slack-aware	46	90.9 [90.0, 91.7]	5.46 [4.38, 6.65]
Fleet-matched	46	83.0 [81.9, 84.0]	10.47 [8.29, 12.87]
Original procedure	46	96.6 [95.1, 97.9]	2.20 [1.26, 3.31]
Screened conservative	46	97.1 [96.3, 97.8]	2.20 [1.47, 2.99]
Capped procedure	46	97.7 [96.4, 98.9]	0.91 [0.56, 1.32]
Clock approximation	44	88.0 [87.2, 88.7]	6.73 [5.39, 8.23]

Table 17 Absolute planned resource means on the same subset as Table 16: distance in km, scheduled duration in minutes, and vehicles. These are not optimality claims.

Configuration	k	Distance	Duration	Fleet
Nominal	46	1180.7 [989.9, 1381.0]	2906.5 [2379.9, 3481.3]	7.67 [6.46, 8.98]
Conservative speed	46	1216.2 [1021.5, 1422.0]	3352.5 [2738.5, 4011.0]	8.30 [6.93, 9.76]
Slack-aware	46	1196.3 [1003.0, 1404.0]	2902.0 [2380.3, 3466.2]	7.46 [6.22, 8.78]
Fleet-matched	46	1174.2 [989.2, 1372.8]	2833.6 [2330.6, 3382.6]	7.46 [6.22, 8.76]
Original procedure	46	1247.9 [1049.3, 1453.0]	3171.2 [2620.9, 3777.3]	8.17 [6.89, 9.57]
Screened conservative	46	1252.8 [1056.4, 1461.2]	3494.5 [2874.0, 4161.0]	8.76 [7.35, 10.28]
Capped procedure	46	1283.5 [1076.5, 1506.8]	3317.3 [2709.4, 3964.1]	8.52 [7.13, 10.04]
Clock approximation	44	1137.2 [955.5, 1330.9]	2868.3 [2335.0, 3456.8]	7.64 [6.32, 9.11]

Table 18 Service-threshold sensitivity on the common 46; columns use the same plans and fresh primary bank. At $n = 20/50/100/200$, allowed-late counts are 2/5/10/20 at $\kappa = .90$, 1/2/5/10 at .95, 0/1/2/4 at .98, and zero at 1.00.

κ	N	C	S	P	SC
0.90	38.8 [36.5, 41.2]	80.6 [78.8, 82.4]	68.8 [65.9, 71.7]	89.2 [84.2, 93.7]	91.2 [88.5, 93.8]
0.95	21.9 [20.3, 23.5]	65.6 [63.1, 68.0]	52.8 [49.9, 55.8]	83.1 [76.3, 89.4]	82.8 [77.9, 87.6]
0.98	11.2 [10.3, 12.2]	48.2 [45.6, 50.8]	36.4 [33.8, 39.0]	75.9 [68.2, 83.3]	73.2 [67.1, 79.3]
1.00	6.4 [5.4, 7.4]	34.4 [31.8, 37.0]	26.1 [23.8, 28.5]	68.2 [59.6, 76.2]	62.4 [53.8, 70.9]

B.1 Instance-level failure audit

Table 19 Every original screening failure. Best buffer and lower confidence bound (LCB) refer to a returned candidate; first no-plan is an observed solver outcome; first proof is the earliest grid point with a reachability certificate. Unresolved indicates at least one no-plan result without that certificate. A dash means no certificate through 60 minutes.

n	Seed	Best β	LCB (%)	No-plan β	Proof β	Unresolved
20	7200	45	90.7	50	–	Yes
20	7201	35	93.5	40	–	Yes
20	7210	35	90.8	45	–	Yes
20	7212	5	37.9	10	–	Yes
20	7216	55	90.7	60	60	No
20	7218	10	65.9	15	50	Yes
50	7501	25	87.5	30	35	Yes
50	7503	10	64.1	15	15	No
50	7504	35	91.3	40	40	No
50	7506	0	44.7	5	5	No
50	7508	35	85.9	40	40	No
50	7511	35	88.5	40	50	Yes
50	7513	40	93.6	45	60	Yes
100	8003	20	85.3	25	25	No
100	8004	15	81.5	20	20	No
100	8007	5	73.6	15	15	No
100	8008	25	91.7	35	35	No
200	9004	10	71.2	15	15	No
200	9005	20	88.9	25	25	No
200	9006	25	94.9	40	40	No
200	9007	20	86.4	25	25	No

Table 20 Retained equal-budget deterministic-core benchmark: OR-Tools excess distance relative to PyVRP/HGS (%). Counts are valid one-second/five-second pairs. There was one missing pair at the one-second budget.

n	Valid pairs (1s/5s)	One second	Five seconds
20	19/20	1.0 [0.3, 1.8]	0.0 [0.0, 0.0]
50	15/15	4.8 [3.3, 6.2]	3.0 [1.9, 4.2]
100	10/10	8.5 [5.9, 11.1]	6.3 [4.7, 7.9]
200	8/8	32.9 [29.4, 36.2]	11.1 [9.3, 13.5]

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